To begin with, I would like to apologize for taking so long to return this review. However, I had some real issues with the paper, and rather than return an "old man shouting at clouds" negative review, I wanted to take the time to show the authors how I think that the study should have been conducted.

Andreotti et al. present a modification of tree-based birth-death sampling approaches that focus on occurrence data. The goal is straightforward: to estimate the probability of occurrence distributions in the fossil records in order to estimate the likelihoods of diversification and sampling rates over time. They do this using cetacean data, which represents a very good "model" system for this sort of study because whales have both extensive phylogenetic analyses conducted by myriad workers over the last 30+ years and one of the most completely *entered* fossil records in the Paleobiology Database thanks to whale specialists (particularly Mark Uhen) who have been involved with the PBDB for (now!) decades. In addition to the occurrence data, this also means that the PBDB has extensive taxonomic information so that obsolete genus-species combinations and junior synonyms (as well as reidentifications) are connected to the appropriate OTUs.

I have two big issues with this paper. One, the sampling models are naïve and (in some ways) backwards: better ones have been used before and probably should be used here. In particular, the separation of sampling rates for coded and uncoded taxa is not justifiable. Two, the paper ignores multiple decades of relevant paleobiological work. This turns up in multiple ways: a) the authors give credit to concepts and approaches developed by paleobiologists years (even decades) ago to much more recent (and usually non-paleontologist) authors; b) the basic approach itself replicates or nearly replicates methods developed in the paleontological literature years ago without any reference to them; and, c) the authors seem to know very little about macroevolutionary theory developed largely by paleontologists that probably provides better explanation for some basic patterns and (which in turn) have roots in generally accepted ecological and macroecological theory. A more minor issue that I have is that the authors' terminology often is hard to follow and (I suspect) different from the standard terms used in paleontology and ecology. Now, if the authors' goal is to alienate and anger paleontologists, then this paper will do that (and, again: it's part of why it's taken me so long to complete this review!): but I really doubt that is their intent. However, given that the Paleobiology Database was founded nearly 30 years ago in order to allow paleontologists to take into account variable sampling over time when doing evolutionary studies, it seems almost rude to be using occurrences from that database while pretending that the paleontologists entering those data never thought of these issues.

Sampling and sampling distributions. I will begin by addressing how the authors deal with sampling. I will note that some of these issues transcend this paper. The abstract opens with a fairly audacious line:

> "*...we develop the Occurrence Birth-Death Diffusion (OBDD) process. This process incorporates the sampling of fossil occurrences by extending the recently developed Fossilized Birth-Death Diffusion (FBDD) process, thereby taking advantage of a clade's entire fossil record to infer its evolutionary history.*"

So, what does the clade's fossil record look like? Now, first I will point out that "entire fossil record" is a poor choice of words because no matter how much someone includes, somebody will be able to point out some aspect of the fossil record that you are excluding. However, I am going to focus on four very basic aspects of a clade's fossil record that this method should (but fails) to accommodate. This centers around the "main" equation of the paper (eq. 0.2):

$$P[\Omega|\Psi,\omega] = \prod_{i=1}^{I} \frac{(\omega L_i)^{k_i} e^{-\omega L_i}}{k_i!}$$

where L is the lineage duration, Ψ is the phylogenetic topology, ω is sampling, k_i is the number of fossil occurrences associated with lineage I and Ω are the occurrence data. (On a side note, the usual custom would be to make this K, as this is not an array of sampling rates, but an array of occurrences.) The authors should note here that this is a variant on Huelsenbeck & Rannala's (1997 *Paleobiol.* 23:174) equation for estimating topology likelihood given stratigraphic data. Even closer to what the authors are doing, this equation is, for all intents and purposes, identical to Starrfelt & Liow's (2016 *Phil. Trans. Royal Soc. London Ser. B. Biol. Sci.* 371:20150219) TRiPS ("True Richness estimated using a Poisson Sampling model) method. The sole difference is that by doing this in a phylogenetic context, "ghost lineage/taxon" range extensions (see Smith 1988 *Palaeont.* 31:799; Norell 1993 *Am. J. Sci.* 293–A:407) with zero finds are explicitly included. TRiPS estimates the number of lineages with zero finds in a manner similar to Chao-2 or other richness estimators (see, e.g., Chao 1987 *Biometrics* 43:783): given ω, the probability of zero finds gives us the expected proportion of unsampled taxa, so it also takes into account unsampled lineages. (I have more to say on this below.)

The one twist is that the authors add is that they put a Gamma distribution on the sampling rates within individual intervals. However: sampling rates among contemporaneous taxa do not follow Gamma distributions or even Beta distributions (the analog for discrete data, which occurrence data are: see below): instead, occurrence/occupancy data for fossil taxa follow lognormal distributions

(Wagner & Marcot 2013 *MEE* 4:703; Foote 2016 *Paleobiol.* 42:707; note that subsequent analyses explicitly using Gammas as well as Betas still find that lognormals do the best jobs of describing the distributions of occurrences in PBDB data). Now, this probably won't make a huge difference: but why not use a distribution with empirical backing?

Coded vs. Uncoded Taxa Sampling Distributions. The authors add one other strange and (in my opinion) inappropriate twist to this: they use one sampling rate for coded taxa and another for uncoded taxa. (At least this is what I think that they mean by "phylogenetic fossil" vs. "fossil occurrence" on lines 44-5 on page 3). This is simply backwards. Considering 2+ sampling groups is appropriate if two groups might represent different sampling regimes: e.g., different environmental preferences, different taphonomies, different biogeographic distributions, etc. (see, e.g., Wagner & Marcot 2013 *MEE* 4:703; Finarelli & Liow 2016 *Biol. J. Linn. Soc.* 118:26; Franeck & Liow 2019 *Paleobiol.* 45:221). That is not the case for coded vs. uncoded whales. Figure 1 shows the distribution of finds for the OTUs analyzed in this study based on PBDB data as of November 2025. I used the matrices in the supplementary information to determine which taxa are coded and which are not; note that the three most commonly found uncoded taxa might actually be coded in one matrix or another, but it was somewhat ambiguous, so I tallied them as uncoded. Unsurprisingly, the uncoded taxa are more apt to come from those species known from fewer collections. This does not represent a separate sampling distribution: there are no taphonomic differences between these taxa and others, and as they commonly co-occur with coded taxa, they do not

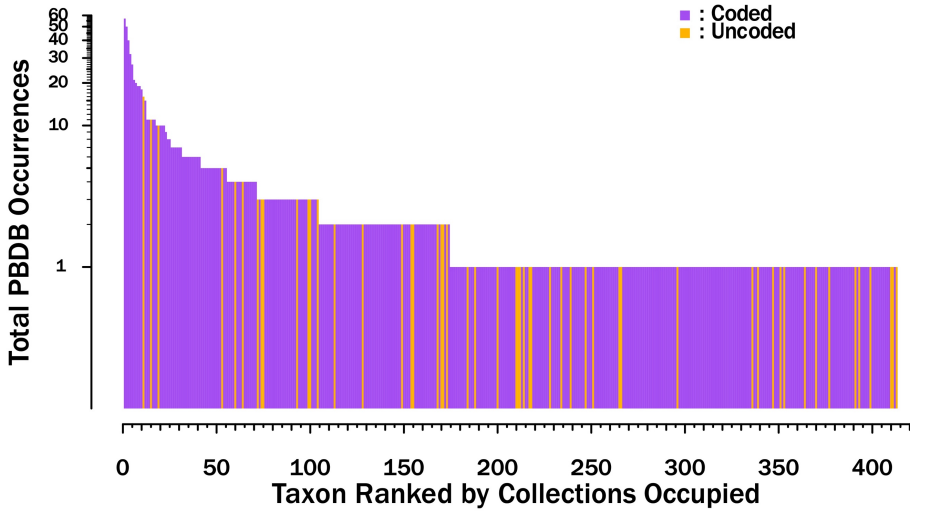

Figure 1. Occurrence distributions for cetacean OTUs used in authors' study. Extant & extinct taxa are separated. Note that Holocene occurrences are excluded. Some OTUs (particularly unnamed taxa given only as specimen numbers) could not be found in the PBDB.

represent separate environmental or biogeographic realms. Instead, this reflects the fact that the workers are most apt to code the species with lots of specimens. There most obvious reason for this is that these are the species that workers are most apt to find easily when exploring museum collections. In particular, those species known from 10s of localities are going to have specimens in lots of museums, and they usually will be the first ones to get coded. It is also much less probable that the only "good" specimens will be in smaller, less easy-to-visit collections when species are known from multiple localities. And, of course, the more specimens that are available, the higher the probability that specimens worth coding (or that plainly show differences separating a species from related species) is found.

In other words, the actual "function" here is the probability that a taxon will be coded given that it is found from 1…n localities. That might be of interest for some types of studies, but it is not here because the completely unsampled whale species are generally going to come from the tail of low sampling rates *among all whales/marine tetrapods/marine vertebrates*.

Perhaps the best way to summarize the inappropriateness of separating sampling between coded and uncoded taxa comes from examining the sampling implicit to the distribution of taxa known from 1, 2, 3, etc., sites shown in purple in Figure 1. A simple Chao2 estimator (Chao 1987 *Biometrics* 43:783) infers that we have sampled a bit over half of the coded taxa. That is, of course, logically impossible: coded taxa necessarily come from the subset of taxa that are sampled.

Now, there are cases where there might be sampling rates will vary for reasons that we can identify a priori differences in sampling. For example, Foote et al. (2024 *Biol. Lett.* 20:20230475) recently contrasted logistic diversity patterns among epifaunal and infaunal bivalves using a birth-death-sampling model that allowed for different sampling rates among epifaunal and infaunal clams. This sort of difference might be observable even within the restricted study systems of typical phylogenetic analyses. Taphonomic factors such as skeletonization vs non-skeletonization, aragonitic vs. calcitic skeletonization, etc. usually vary on taxonomic scales much larger than those at which we conduct phylogenetic analyses, but many studies of Cambrian taxa would include OTUs of different preservation potential due to differences in the extent of carapace mineralization. Here, marine vs. non-marine would apply *if* the analysis was on all artiodactyls including whales.

<u>Expected Sampling Can and Should Be Assessed in Advance</u>. As I note above, sampling is *not* a Poisson process: it is a binomial (or multinomial) one dependent

upon the number of sites/collections. If Interval A has 100 relevant collections, then the maximum number of occurrences is 100. This might seem obvious, but it is by far the single biggest factor affecting expected sampling within an interval. Suppose the next Interval B has 50 collections. The null sampling model under the Poisson model is sampling rate ω for both intervals. In reality, it would be $\omega\times100$ finds for Interval A and $\omega\times50$ finds for Interval B, where ω now represents the per-site probability of finding a species. (Under a distribution such as a lognormal, it would be $\omega_i\times100$ finds and $\omega_i\times50$ for A and B respectively). Under a Poisson model, this almost certainly would call for a sampling rate shift, adding complexity to the model. However, under a binomial model, we would accept rate shifts only if the sampling rates per locality are sufficiently more likely that it's preferable. This was, after all, why we started the Paleobiology Database all of those years ago: to accommodate variable sampling opportunities over time.

So, how many sampling opportunities do we have for whales? We now need to consider a very important concept that never is mentioned in this paper: taphonomic controls, i.e., localities where whales *might* be found but are not (Bottjer & Jablonski 1988 *Palaios* 3:540). A clade's "entire fossil record" is not just, say, the 100 whale localities yielding whales in interval A: it includes the localities that *could* have yielded whales but failed to do so. Otherwise, we overestimate the sampling potential of the group (see, e.g., Alroy et al. 2001 *PNAS* 98:6261). Figure 2 shows the PBDB download used to generate Figure 1, with two possible taphonomic controls. Note that the dates are somewhat more refined than output by the PBDB as the upper and lower bounds of localities are modified using an external database based on foram, nannofossil, isotope, polar chrons and/or radiometric dates (see Congreve et al. 2021 *Proc. R. Soc. B* 477:20211450), either directly for particular localities or more generally for the rock-unit at that locality. I then binned them into "stage-slices" that typically represent 2-3 million year intervals. I include two levels of taphonomic controls that might seem appropriate: (the Sirenia, Desmostylia, Pinnipedia and marine Crocodylia), and marine

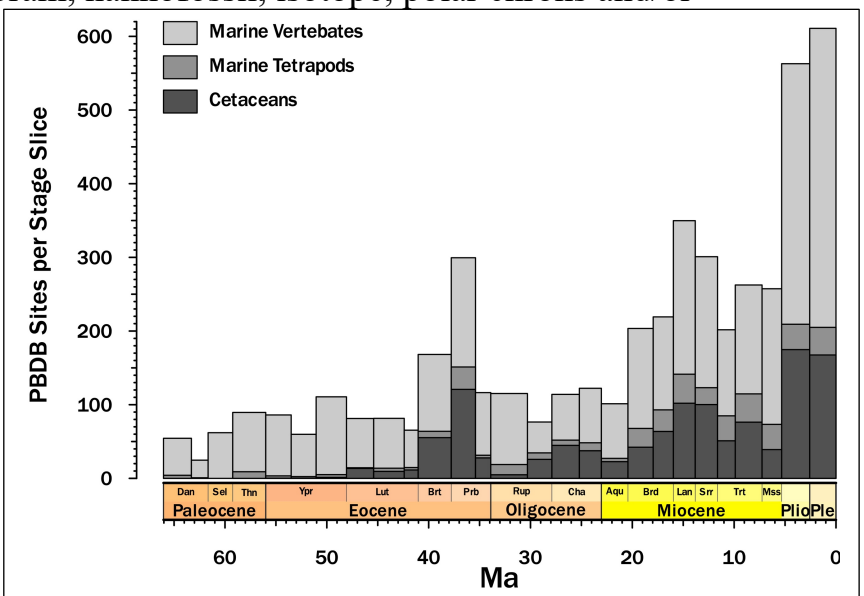

Figure 2. Relevant fossiliferous collections over time from the Paleobiology Database. Holocene data are omitted.

vertebrates in general. In both cases, only sites in which species can be identified (rather than just genera) are included. Marine tetrapods might seem more appropriate because these fossils are more similar to whale bones. However, these control sites also are very reliant on the diversity of marine tetrapods, which waxes & wanes enormously over the Cenozoic. Many times, whales are the only major group. Moreover, there are almost no recognized opportunities to find whales before they evolve under this control. Therefore, it probably is more appropriate to use vertebrates as a whole: if you can find these elements in a particular facies, then whale elements should have a non-zero chance of being preserved if whales were present. (There are no major shifts in mineralogy or other taphonomic differences across vertebrates either, and the smaller sizes of most fishes can make them more difficult to find and identify than tetrapods.) Right away this shows us that the total number of sites producing whale bones generally goes up and down with the total number of marine sites preserving vertebrate elements. Moreover, this provides us with important information about the *expected* sampling rates before the earliest known whales. Do this, and you'll probably save considerable search time in these analyses as well as provide an empirical basis for the initial parameterization. You'll also have a simple visual demonstration of the facts that: 1) the quality of the whale record varies through time; and, 2) the primary control on that quality is the larger fossil record, not temporal variation in the sampling of particular whale species. Again, this is really why the PBDB exists in the first place!

(Note that this is not as fine of resolution as the authors get by optimizing uniform distributions of occurrence ages shown in their Appendix Figure C; however, that still could be done after this initial restriction of dates, with provides far finer resolution than the raw dates currently returned by the PBDB. If they really are serious about refining dates further, then I would recommend using quantitative biostratigraphic methods such as Unitary Associations (Guex 1977 *Bulletin de la Societé Vaudoise des Sciences Naturelles* 73:309; for recent examples, see, e.g., Leu et al. 2022 *Swiss Journal of Palaeontology* 141:; Lyu et al. 2023 *Palaeogeog., Palaeoclimat., Palaeoecol.* 627:111721) or Appearance Event Ordination (Alroy 1994 *Paleobiol.* 20:191). Indeed, I would strongly recommend that the authors consider collaborating with quantitative biostratigraphers in the future!)

<u>Phylogenies are not the only or even best way to infer missing taxa</u>. On lines 419-421, the authors state: "*In contrast, the phylogenetic modeling framework, including all models presented here, capitalizes on phylogenetic relationships to infer the existence of unsampled lineages.*" I have two major issues with this. First, all birth-death-sampling methods do this. I mentioned TRiPS above. Other BDS

methods such as three-timers (Alroy 2008 *PNAS* 105:11536), capture-mark-recapture (Connolly & Miller 2001 *Paleobiol.* 27:751), forward & reverse survivorship (Foote 2001 *Paleobiol.* 27:602) all explicitly take into account that there are unsampled lineages when reconstructing richness or estimating diversification rates. Methods for estimating distributions of sampling rates (or occupancy rates) also explicitly do this, as the best-fit distributions include "true" richness as a parameter in the same way that TRiPS does. There are only two differences. One is that phylogenetic range-extensions *specify* unsampled lineages: either unsampled precursors to sample lineages (Norell's ghost lineages) and unsampled ancestors (Norell's ghost taxa), whereas other BDS methods indicate that there are X additional taxa in an interval that could be ghost lineages or ghost taxa, or missing side branches or carry-overs.

The second big and even more important difference is a corollary of the first: the missing taxa specified by phylogenies infer unsampled lineages *in one temporal direction only*. This is a big problem. Stratigraphic ranges underestimate true species durations in both direction: the first occurrence is later than the true origin, and the last appearance precedes the true extinction (i.e., the Signor-Lipps effect; Signor & Lipps 1982 *Geol. Soc. America Spec. Papers* 190:291; note that because the initial focus of the SL effect was on the study of extinction, some workers refer to the difference between last appearance and extinction as "Signor-Lipps Effect" while calling the difference between origination and first appearance the "Sppil-Rognis Effect"). At the clade level, we get both unsampled ancestors and unsampled dead-ends appearing after sampled taxa. As a result, even correct phylogenies can bias phylogenies towards depicting logistic or other diversity dependent diversification patterns (Wagner 1995 *Paleobiol.* 21:410). This can be critical for understanding macroevolutionary dynamics. For example, birth-death-sampling analyses by Foote (2007 *Paleobiol.* 33:261) suggest that extinction might be much more concentrated into particular events than last appearance data alone imply due to the Signor-Lipps effect (but less so originations). Because there is no way for phylogenies to reconstruct "Signor Lipps Lineages" (sensu Wagner 2000 *Syst. Biol.* 49:65), there is no way that the approach offered here could suggest this. Now, I've been a big advocate of tree-based approaches since the 1990s, particularly for analyzing patterns of anatomical evolution. However, when it comes to general clade-wide diversification dynamics, we are better off skipping the phylogeny and simply using birth-death-sampling methods derived from population biology. That stated, these sorts of approaches will be important is when we are trying to get at whether rates of diversification or rates of anatomical change differ on different parts of a tree; in these cases, we'll

want the best possible use of occurrence data in order to come up with the most realistic branch durations and relationships for evaluating these ideas.

Why use whales? At some point, the authors really should explain why cetaceans are a good model group for this study. I would simply note that not only do cetaceans have a good fossil record, but that it is very well represented in the Paleobiology Database thanks largely to the efforts of Mark Uhen. Over one third of the taxa analyzed by the authors are known from 2+ localities, with over one fifth of those species crossing at least on chronostratigraphic boundary and over one sixth crossing 2+ chronostratigraphic boundaries and thus having minimum stratigraphic ranges >0. Finally (although the authors do not note this!), the attention from whale specialists such as Uhen means that the PBDB has extensive taxonomic information about those whale species, meaning that junior synonyms, obsolete combinations and even re-identifications. In addition to this, there has been extensive phylogenetic study of whales by Uhen and several other workers. This results in well over 400 different taxa (nearly all of which have occurrences in the PBDB, including unnamed museum specimens) having been coded for at least some anatomical characters.

More minor comments:
Pg. 3 lines 44-45. *From now on, we will use "fossil" or "phylogenetic fossil" to designate those in the tree and "occurrence" or "fossil occurrence" for those that are not.*

As I note above, this is inappropriate. Moreover, this terminology is awful and confusing. A fossil occurrence means that a species is known from a fossil at a fossiliferous locality. That is the way we've used the term for decades. It applies to any and all taxa. Trying to change it just makes this confusing. What I think that the authors mean is that they are separating coded taxa from uncoded taxa. Just call them that. (And if that is not what the authors' meant, then it needs to be completely rewritten.)

Pg. 3 lines 52-52: *Analyses of the OBD model in the context of pathogen phylodynamics have shown that sampling times (incidence data) often play a key role in driving inference when sequence diversity is low.*

"Incidence" is synonymous with "occurrence" in macroecology and paleobiology (although paleontologists tend to favor occurrence). It's not just a time: it's also a place and often an assemblage of co-occurring taxa. Now, the latter two might be of less importance in virology and epidemiology, but they are very important to the

concept of incidence/occurrence with which most of the reading audience will be familiar.

Pg. 4, lines 71-73. *Substantial progress has also been made in developing diversification models that accommodate rate heterogeneity, in which rates can differ between portions of the tree, either due to rare clade-wide shifts*

Maybe the authors mean in the context of phylogenies: but with general data, paleontologists have been using models that demonstrate rate heterogeneity for over 40 years! For example, the work showing that the classic "Three Faunas" are typified by different diversification rates is now over 40 years old (e.g., Sepkoski 1981 *Paleobiol.* 7:36). This really needs to be pointed out: too often these sorts of papers act as if paleontologists haven't done anything when we've done a ton.

Pg. 5 lines 96-100. "*..., Quintero et al. (2024) observed an imbalanced speciation process in mammals wherein most lineages undergo a progressive decline in speciation rates before going extinct - notably at the K-Pg mass extinction event - while a few lineages preserve a high speciation potential and drive the diversification of their clade.*"

Quintero et al. almost certainly independently rediscovered that mammals show logistic diversification, in which origination rates depend on standing richness. This was originally shown for mammals 30 years ago (Alroy 1996 *Palaeogeog., Palaeoclimat., Palaeoecol.* 127:285). Somehow, those authors failed to even discuss diversity-dependent models, despite the fact that they've been widely discussed in the paleontological literature for over 40 years and despite the fact that the basic model has been corroborated in far more studies than I can cite here. As a result, they settled on this an interpretation that lacks any basis in macroevolutionary theory. This is a big reason why I strongly encourage the authors to read the macroevolutionary literature: these sorts of mistakes will encourage paleontologists and other macroevolutionary biologists to not take their other work seriously.

(It seems that Quintero et al. did consider logistic diversification briefly in their 2025 paper but they dismissed it because the diversity curve looked incorrect; however, it has long been known that richness over time can fluctuate substantially because of variation in extinction rates and that the appropriate tests contrast how well richness predicts origination rates.)

Pg. 5, lines 102-105: "*Quintero et al. (2025) studied diversity trajectories across 27 radiations and used data-augmented histories to show that diversity expansions are fueled by bursts of rapid speciation, while slowdowns and declines are primarily governed by extinction rates.*"

This was first demonstrated nearly 40 years ago by Gilinsky & Bambach (1987 *Paleobiol.* 13:427)! That said, slowdowns usually are governed by reduced origination rates, as nearly all groups show logistic diversification patterns in which origination rates decline as richness increases while extinction rates and equilibrial origination rates tend to fluctuate slightly together save in elevated extinction events.

Pg. 5, lines 106-108: "*paleontological databases, in which morphospecies are often only represented by a single described specimen (Wright et al., 2022).*"

This statement is true, but I feel that it is misleading. Almost exactly 50% (49.9%!) of the OTUS in the whale matrices used for this study include 1+ polymorphic character. I will not state that it is impossible to recognize polymorphisms without 2+ specimens, but I am hard-pressed to think of how we might do that. In fact, polymorphisms are sufficiently ubiquitous with paleontological data that studies have been made concerning temporal patterns of polymorphisms within trilobites (Webster 2007 *Science* 317:499) and paleontologists have been especially concerned with the effects of polymorphisms on phylogenetic analyses (e.g., Parins-Fukuchi 2024 *Paleobiol.* 50:17). Now, this doesn't change the fact that the single most common occurrence number for species is 1 (see, e.g., Buzas et al. 1982 *Paleobiol.* 8:143): but even species with single occurrences might be known from multiple specimens from that single site. Moreover, and as Figure 1 corroborates, there probably is a bias towards coding species known from 2+ specimens. Regardless, the authors should phrase this in a way to make it clear that single-specimen taxa are not the overwhelming majority of analyzed taxa in most phylogenetic studies.

Pg. 7 lines 145-146. "*145 only recorded as sampling times (as shown in Fig. 1). Given that fossilization and fossil recovery rates are expected to vary over geological time (Dominici et al., 2020)…*"

Wow, did it really take us until 2020 to realize that? Seriously, I have no idea what the oldest reference to this is: I always go with Raup (1975 *Paleobiol.* 1:333), as this was the first paper to recommend rarifying finds from different intervals down to common sampling levels in order to "sampling standardize" richness estimates.

However, I think Darwin commented on the fact that sampling varied over time. At any rate, and to repeat myself, the PBDB was founded in 1998 explicitly to deal with temporal, geographical and environmental heterogeneities in the fossil record. At this point, that is a long time ago, which makes this a very old idea worthy of a very old reference.

(And, again: if I wanted to write a line to anger paleontologists, then I would have been hard pressed to come up with a better one that this.)

Pg. 8 lines 158-159. *Therefore, a complete phylogeny, combining the reconstructed tree and the augmented branches,…"*

I know that I am getting repetitive, but this "complete" phylogeny still omits many branches: "Signor-Lipps" lineages and completely unsampled non-ancestral taxa all are excluded. So, it really is not complete at all.

Pg. 9 lines 185-186: *The ~~likelihood~~* ***probability*** *of the whole occurrence dataset Ω given the complete augmented tree Ψ and the occurrence sampling rate…*":

It is the tree and the sampling rates that have likelihoods, not the data. Thus, it should state "The probability of the whole occurrence dataset…" or "The likelihood of the tree and sampling rates". Along these lines, eq. 0.2 should be either:

$$P[\Omega|\Psi,\omega] = \prod_{i=1}^{I} \frac{(\omega L_i)^{k_i} e^{-\omega L_i}}{k_i!}$$

or

$$L[\Psi,\omega|\Omega] = \prod_{i=1}^{I} \frac{(\omega L_i)^{k_i} e^{-\omega L_i}}{k_i!}$$

Pg. 12 lines 253-255: "*We used the cetacean phylogeny from Lloyd and Slater (2021), which contains 89 extant species and 405 fossils (representing distinct morphotaxa).*"

I think that the authors mean to write that they used 405 fossil taxa representing distinct morphotaxa (or morphotypes). Those 405 fossil taxa almost certainly represent many more than 405 fossils or even 405 individuals, particularly given how common polymorphic taxa requiring 2+ specimens are in these data.

Pg. 12. Lines 259-260: *In addition, we used the 966 fossil occurrences from the Paleobiology Database (McClennen, 2024) used in Andréoletti et al. (2022).*

The appropriate reference is Uhen et al. (2023 *PaleoBios* 40:1), which outlines how to use the PBDB and best practices. People often cite Alroy et al. (2001 *PNAS* 98:6261) as this was the first "big" PBDB paper. One of those best practices is to cite the 10 papers contributing the greatest numbers of occurrences, and to recognize the top 5+ contributors in the acknowledgements. The authors should also get a PBDB publication number from the PBDB secretary (currently Noel Heim) once the paper is finally accepted.

(Mark McClennen is a wonderful person, but he's the programmer, not the designer.)

Pg 12 line 264: *"We fitted eight diversification models, all implemented in Tapestree.jl: "*

These are not really diversification models, at least in my book. What I would consider to be diversification models would be exponential vs. logistic vs. hierarchical diversification (see Brayard et al. 2009 *Science* 325:1118), with alternative models of peaks in extinction rates (e.g., coinciding with the PETM, end-Paleogene, etc.), with simple exponential growth and constant extinction being the null model.

Pg. 13, lines 279-282*: ...enabling rate changes at the start and end of the Early Oligocene (Rupelian), Early Miocene (Aquitalian), and Late Miocene (Messinian) periods of well-established low preservation (Marx et al., 2016; Dominici et al., 2020)"*

Or you could just use Figure 2 that I've provided!

Also, I would rephrase the intervals. The Rupelian is equivalent to the Early Oligocene. However, the Aquitanian and the Messinian are only parts of the Early Miocene and Late Miocene, respectively. So, write "Aquitanian (Early Miocene) and Messinian (Late Miocene)" as more of your readers will know when the Miocene is than when the Aquitanian or Messinian. (As a Paleozoic worker, I know these things only because of collaborations with Cenozoic workers!)

Pg. 19. Line 361*: "Occurrence sampling rates are higher than those of phylogenetic fossils, as expected…"*

Again, that's just because we code well-sampled species before we code poorly-sampled ones. And, again: there really should not be two rates here, as this is an expected outcome from having a lognormal or Beta or exponential distribution of sampling intensities/occupancy rates among species.

Pg. 19. Line 365 *"...particularly in the Rupelian (33,9-28,1 Ma)."*

The Rupelian is 33.9 – 27.29 in Gradstein et al. (2020 *Geologic Time Scale 2020*), 33.9-27.30 at https://stratigraphy.org/chart/ and 33.9-27.82 at https://rock.geosociety.org/net/documents/gsa/timescale/timescl.pdf?v=2022. What is the source for the 28.1?

Pg. 23 lines 408-410: *These alternative occurrences could include teeth, scales, plant organs (leaves, seeds, pollen), biochemical proxies, ichnofossils (trace fossils), coprolites.*

Teeth are not a great example, at least for mammals, as they figure very heavily in phylogenetic analyses of mammals. Indeed, mammal teeth are much more diagnostic of species and genera than are post-cranial material: whereas closely related mammal species (e.g., humans and chimps) can be distinguished by teeth (even if they are very similar!), variation in post-cranial skeletons often reflects only allometry. Cranial material is somewhere in between it seems: for example, most of the variation in skulls among the other great apes reflects most allometry. For groups such as dinosaurs, teeth diagnose much more general groups: they'll tell us that a family was present, but not whether more than one species of that family was present. Perhaps the more important point here would be in reference to my Figure 2: if there are (say) no Early Barremian localities with anything other than terrestrial tetrapod teeth, then there are zero opportunities to sample codable taxa from that interval. That in turn means that $\omega_{eBar}N=0$ no matter what ω_{eBar} is because N (number of taphonomically controlled sites) = 0.

Similarly, paleobotanists try to use all of the plant organs that they can find. A real issue for them is ascertaining which plant parts came from the same species, but that's a separate issue.

We run into different issues with biochemical proxies. These rarely are closely tied to particular taxa. Moreover, there often is more than one way to get the same biochemical signal. This has come up with reference to the origin of sponges, as biochemical markers typical of sponges appear early in the Neoproterozoic;

however, there are other possible explanations for those markers (see Shawar et al. 2025 *PNAS* 122:e2503009122), which makes the situation tricky. It's definitely evidence that sponges were present: but it's not proof that they were.

Trace fossils and coprolites run into similar problems as non-mammalian teeth: they diagnose very large groups phylogenetically, and they might show you that a general group was present, but probably a group more general than the clade being analyzed.

Pg. 23. Lines 419 – 421: *In contrast, the phylogenetic modeling framework, including all models presented here, capitalizes on phylogenetic relationships to infer the existence of unsampled lineages.*

As I note above, all birth-death-sampling methods do this. Moreover, they are agnostic as to whether the inferred unsampled lineages are younger than, older than or contemporaneous with sampled taxa.

Pg. 23. Line 424 *Heterogeneous rates give us a refined view of a clade's evolution*

Again, paleontologists have known this for decades.

Pg. 24. Lines 435-437: *At the individual lineage level, we observe a general decline in net-diversification rates (similar to the "insidious loss of species macroevolutionary fitness" described in Quintero et al. (2025)), except for a few lineages."*

Again, this is almost certainly picking up logistic diversification. This should be possible to model, if you can provide RevBayes with information about estimated richness for intervals This actually could be gleaned from the distribution of occurrences within an interval using something as simple as a Chao-2 estimator, which is remarkably robust to differences in occurrence distributions. However, you'd have to add code to RevBayes to do it. Moreover, it would change as ω changes, as we expect more unsampled taxa limiting the diversification potential of the entire clade as ω decreases.

If the authors have any questions about what this review or interest in using the stratigraphic databases used to refine PBDB occurrences, then I would be happy to discuss it with them. Now, I realize that this review will come across as very

negative, and it is entirely probable that the authors will be more interested in buying me a ticket on a rocket destined to the Sun than discuss this work with me! However, I was not joking about this paper being offensive to paleontologists: I know that there schools out there that assume that we've never done anything quantitative, this simply is not the case, and the authors are venturing into issues for which we have been devising probability and likelihood models for many years. They are not perfect, but they also are not worthless. At any rate, I can reached at:

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